

Problem

An elastic ball bounces on a horizontal plate in a gravitational field g . The value of g changes slowly over time. Determine how the bounce height H depends on g .

Solution

1. Energy of the ball

Let the vertical coordinate x be measured upward from the plate. Between bounces, the ball moves freely under gravity. Its total mechanical energy is

$$E = \frac{m\dot{x}^2}{2} + mgx,$$

where $g = g(t)$ varies slowly.

2. Time derivative of the energy

Differentiating with respect to time:

$$\dot{E} = m\dot{x}\ddot{x} + \dot{m}gx + mg\dot{x} + m\dot{g}x.$$

Since m is constant, $\dot{m} = 0$. Using the equation of motion $\ddot{x} = -g$, we obtain

$$\dot{E} = m\dot{x}(-g) + mg\dot{x} + m\dot{g}x = m\dot{g}x.$$

3. Averaging over a bounce period

Because g changes slowly, we average $\dot{E}$ over one full flight period T :

$$\langle \dot{E} \rangle = \frac{\int_0^T m\dot{g}x \, dt}{T}.$$

For an elastic bounce, the speed just after impact is V_0 , and the flight time is

$$T = \frac{2V_0}{g}.$$

Setting $x = 0$ at lift-off ($t = 0$), the coordinate is

$$x(t) = V_0t - \frac{gt^2}{2}, \quad 0 \leq t \leq T.$$

4. Calculating the average

Substituting $x(t)$ and T :

$$\langle \dot{E} \rangle = m\dot{g} \cdot \frac{\int_0^{2V_0/g} \left(V_0 t - \frac{gt^2}{2} \right) dt}{2V_0/g}.$$

Compute the integral:

$$\begin{aligned} \int_0^{2V_0/g} V_0 t dt &= V_0 \cdot \frac{1}{2} \left(\frac{2V_0}{g} \right)^2 = \frac{2V_0^3}{g^2}, \\ \int_0^{2V_0/g} \frac{gt^2}{2} dt &= \frac{g}{2} \cdot \frac{1}{3} \left(\frac{2V_0}{g} \right)^3 = \frac{g}{2} \cdot \frac{8V_0^3}{3g^3} = \frac{4V_0^3}{3g^2}. \end{aligned}$$

Thus,

$$\int_0^{2V_0/g} \left(V_0 t - \frac{gt^2}{2} \right) dt = \frac{2V_0^3}{g^2} - \frac{4V_0^3}{3g^2} = \frac{2V_0^3}{3g^2}.$$

Therefore,

$$\langle \dot{E} \rangle = m\dot{g} \cdot \frac{2V_0^3/(3g^2)}{2V_0/g} = m\dot{g} \cdot \frac{V_0^2}{3g}.$$

5. Expressing in terms of energy

At lift-off, the energy is purely kinetic:

$$E = \frac{mV_0^2}{2} \quad \Rightarrow \quad V_0^2 = \frac{2E}{m}.$$

Substituting:

$$\langle \dot{E} \rangle = m\dot{g} \cdot \frac{2E/m}{3g} = \frac{2}{3} \frac{\dot{g}}{g} E.$$

6. Adiabatic invariant

Treating E as the slowly varying average energy, we have

$$\frac{dE}{dt} = \frac{2}{3} \frac{\dot{g}}{g} E \quad \Rightarrow \quad \frac{dE}{E} = \frac{2}{3} \frac{dg}{g}.$$

Integrating:

$$\ln E = \frac{2}{3} \ln g + \text{const} \quad \Rightarrow \quad E \propto g^{2/3}.$$

7. Finding the bounce height

The maximum height H satisfies $E = mgH$. Hence,

$$mgH \propto g^{2/3} \quad \Rightarrow \quad H \propto \frac{g^{2/3}}{g} = g^{-1/3}.$$

Answer

$$H \propto g^{-1/3}$$

If gravity increases slowly, the bounce height decreases as $g^{-1/3}$; if gravity decreases slowly, the bounce height increases.